

Shubhojyoti Datta Chaudhuri

2002datta@gmail.com | +1 (716) 235-6598 | Buffalo, NY | [linkedin.com/in/shubhojyotidatta](https://www.linkedin.com/in/shubhojyotidatta) | shub-portfolio-xi.vercel.app

SUMMARY

Computer Science graduate student focused on software engineering, AI/ML systems, and data-driven applications. Hands-on experience building Django REST APIs, RAG pipelines, distributed data processing workflows, reinforcement learning environments, and full-stack ML applications. Skilled in Python, C/C++, JavaScript, SQL, Docker, LangChain, PySpark, and cloud-backed application development.

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, SQL, HTML, CSS

Backend & Frameworks: Django, Django REST Framework, Flask, React, REST APIs, Flutter

AI / ML: LangChain, RAG, FAISS, Gemini, PyTorch, Hugging Face Transformers, Stable-Baselines3, PPO

Data & Distributed Systems: PySpark, Hadoop, HDFS, Feature Engineering, Model Evaluation

Databases & Tools: MySQL, PostgreSQL, Firebase, AWS RDS, Docker, Git, Figma, Postman

EDUCATION

University at Buffalo, The State University of New York

Buffalo, NY

Master of Science in Computer Science, AI/ML Track; GPA: 3.7/4.0

Aug 2025 – Dec 2026

Chandigarh University

India

Bachelor of Engineering in Computer Science; GPA: 3.2/4.0

Aug 2021 – Jul 2025

EXPERIENCE

LookThePart

Remote

Graduate Student Developer, CSE 611 Project

Jan 2026 – June 2026

- Designed and developed backend features for a Django-based fashion discovery platform that lets users explore and shop outfits from TV shows and movies.
- Built 50+ RESTful APIs using Django REST Framework for authentication, product search, post sharing, media handling, and mobile app integration.
- Created and maintained 15+ relational database models including Entertainment, Characters, Products, Posts, and media-related entities to support scalable content workflows.
- Integrated AWS S3 and Pillow-based media processing pipelines for image upload, storage, thumbnail generation, and optimized product/post asset delivery.

PROJECTS

Cashback Reconciler – Financial Rewards Reconciliation Engine | *Python, Plaid, LLMs, PyTest*

- Designed and developed a trust-first cashback reconciliation system that matches card-linked offers with posted transactions to automatically detect missing cashback rewards using deterministic reconciliation algorithms.
- Implemented merchant normalization, transaction matching, and issuer reliability scoring with comprehensive unit testing, while scaffolding LLM-based email extraction and Plaid transaction integration for real-world financial workflows.

LookThePart Fashion Discovery Platform | *Django, Django REST Framework, AWS S3, MySQL*

- Developed backend services for a fashion discovery platform enabling users to browse and shop outfits from TV shows and movies through scalable REST APIs.
- Designed relational database models, media processing pipelines, and AWS S3 integration for efficient product, image, and content management across web and mobile applications.

PintOS Thread Scheduler & User Programs | *C, PintOS Kernel, GCC, QEMU*

- Extended the PintOS operating system kernel by implementing priority scheduling, priority donation, process execution, argument passing, and system call support.
- Implemented robust system call handlers and user memory validation mechanisms to safely isolate user programs from kernel execution while improving operating system reliability.

DocChat – RAG-Based Document QA System | *LangChain, Google Gemini, FAISS, Python*

- Built a Retrieval-Augmented Generation (RAG) application enabling users to query PDF documents using natural language through semantic search.
- Integrated LangChain, Google Gemini, and FAISS vector indexing to improve retrieval accuracy while reducing hallucinated responses using structured prompt engineering.

Dodge Survive – Reinforcement Learning Game Agent | *Python, Stable-Baselines3, PPO, Panda3D*

- Developed a top-down survival game with decoupled game logic and rendering, enabling efficient headless reinforcement learning training.
- Built an OpenAI Gym-compatible environment and trained a PPO-based reinforcement learning agent using reward shaping to learn obstacle avoidance through trial-and-error interactions.

Sentiment & Emotion Analyzer | *React, Flask, Hugging Face Transformers, PyTorch*

- Developed a full-stack NLP application that compares predictions from five transformer-based sentiment models and generates ensemble sentiment classification through majority voting.
- Integrated a GoEmotions transformer model for emotion recognition, providing confidence scores, model disagreement detection, and an interactive React-based visualization interface.